

Minecraft Crafting Reference

Complete visual guide to the 3×3 crafting table & furnace. By Walusimbi Leon (SGSS) · Public Domain · Version 1.1

-  Diamond
-  Gold
-  Iron
-  Redstone
-  Planks
-  Cobblestone
-  Obsidian
-  Stick
-  Special Item
-  Food
-  Nether

How to read the grids: Each recipe shows the 3×3 crafting table exactly as you place items. Empty cells (grey) are left blank. Place the shown items in the exact positions, and the result appears in the output slot. Items in **bold colored** cells are what you put in — the color tells you the material type. For *any* tier tool (wood/stone/iron/gold/diamond), just swap the material shown.

1. Tools

Pickaxe

Yield: 1 pickaxe

Diamond	Diamond	Diamond
	Stick	
	Stick	

Tip: First tool worth making in diamond. Substitute diamond with Wood, Stone, Iron, or Gold for lower tiers. Pickaxes are the only tool that can mine stone ores and obsidian.

Axe

Yield: 1 axe

Diamond	Diamond	
Diamond	Stick	
	Stick	

Tip: Chops wood fastest. Also useful as a weapon in a pinch (deals more damage than a sword of the same tier).

Sword

Yield: 1 sword

	Diamond	
	Diamond	
	Stick	

Tip: Swords sweep-attack multiple enemies when fully charged. Diamond sword deals 7 damage.

Shovel

Yield: 1 shovel

	Diamond	
	Stick	
	Stick	

Tip: Digs dirt, sand, gravel, and snow faster than anything else. Also turns grass into grass path.

Hoe

Yield: 1 hoe

Diamond	Diamond	
	Stick	
	Stick	

Tip: Tills dirt into farmland for crops. Also quickly breaks leaves and hay bales. Diamond hoe is a flex.

Flint and Steel

Yield: 1 flint and steel

	Iron	
		Flint

Tip: Used to light fires and activate Nether portals. Always carry one. Flint is obtained by breaking gravel (may take several tries).

Shears

Yield: 1 shears

	Iron	
		Iron

Tip: Collect leaves without destroying them, shear sheep for wool (instead of killing), mine cobwebs, and get string from them. Also harvest honey from beehives.

Fishing Rod

Yield: 1 fishing rod

		Stick
	Stick	String
Stick		

Tip: Fish in any water source. You can catch fish, enchanted books, saddles, name tags, and more. Get string by killing spiders or breaking cobwebs.

Bow

Yield: 1 bow

	Stick	String
Stick		String
	Stick	String

Tip: Ranged weapon. Arrows are crafted separately (flint + stick + feather). Bows can be enchanted with Power, Flame, Punch, and Infinity.

2. Arrow Crafting

Arrow (4)

Yield: 4 arrows

	Flint	
	Stick	
	Feather	

Tip: Arrows are consumed when shooting a bow. Feathers are dropped by chickens. Stack up on arrows before exploring caves or fighting bosses.

3. Armor

Helmet

Yield: 1 helmet • 5 diamonds (or other material)

Diamond	Diamond	Diamond
Diamond		Diamond

Tip: Replace diamond with Iron, Gold, or **Leather** (top row + sides = 5 pieces, leather uses the same shape). Chainmail armor is not craftable — only obtained via trading or loot.

Chestplate

Yield: 1 chestplate • 8 diamonds (or other material)

Diamond		Diamond
Diamond	Diamond	Diamond
Diamond	Diamond	Diamond

Note: Chestplate gives the most protection of any armor piece. Make this first if you're short on materials.

Leggings

Yield: 1 leggings • 7 diamonds (or other material)

Diamond	Diamond	Diamond
	Diamond	
Diamond		Diamond

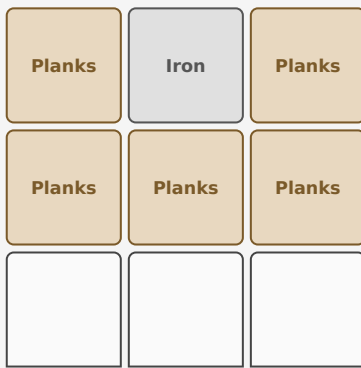
Boots

Yield: 1 boots • 4 diamonds (or other material)

Diamond		Diamond
Diamond		Diamond

Shield

Yield: 1 shield



Tip: Blocks 100% of incoming damage when raised (right-click). Essential for fighting skeletons, creepers, and the Wither. Can be decorated with a banner.

4. Navigation & Utility

Compass

Yield: 1 compass

	Iron	
Iron	Redstone	Iron
	Iron	

Tip: Always points to world spawn, not your bed. If you sleep in a bed away from spawn, the compass still points to world spawn. Use a Lodestone Compass to point elsewhere.

Clock

Yield: 1 clock

	Gold	
Gold	Redstone	Gold
	Gold	

Tip: Shows the position of the sun and moon on its face. Useful underground to know if it's safe to go up. Gimmicky but handy.

Bucket

Yield: 1 bucket

Iron		Iron
	Iron	

Tip: Right-click a water or lava source to collect it. Right-click again to place it. **Pro trick:** Place water at your feet while falling to cancel fall damage. Also collects milk from cows (removes status effects).

Map (Empty)

Yield: 1 empty map

	Paper	

Tip: Hold and right-click to activate the map, then it fills in as you explore. Craft 8 paper around a compass for a zoomed-out map.

5. Machines & Redstone Contraptions

Piston

Yield: 1 piston

Planks	Planks	Planks
Cobble	Iron	Cobble
Cobble	Redstone	Cobble

Tip: Pushes blocks forward when powered. Combine with a Slimeball to make a Sticky Piston (which also pulls blocks back). Foundation of all redstone contraptions: doors, elevators, traps, farms.

Sticky Piston

Yield: 1 sticky piston

	Slimeball	
	Piston	

Tip: Place a Slimeball on top of a Piston in the crafting grid. Also craftable by placing a slime block next to a piston.

Observer

Yield: 1 observer

Cobble	Cobble	Cobble
Redstone	Quartz	Redstone
Cobble	Cobble	Cobble

Tip: The face (with the “U” shape) detects block changes and emits a redstone pulse. Essential for automatic farms — detects when a crop is grown, a piston extends, etc.

Hopper

Yield: 1 hopper

Iron		Iron
Iron	Chest	Iron
	Iron	

Tip: Sucks items from above and feeds them into whatever container is below (chest, furnace, another hopper). Essential for auto-farms, auto-smelters, and sorting systems.

Dispenser

Yield: 1 dispenser

Cobble	Cobble	Cobble
Cobble	Bow	Cobble
Cobble	Redstone	Cobble

Tip: When activated, dispenses items. Arrows get shot, water buckets place water, bonemeal fertilizes crops, armor equips to players. Uses a bow (any durability) in the center.

Dropper

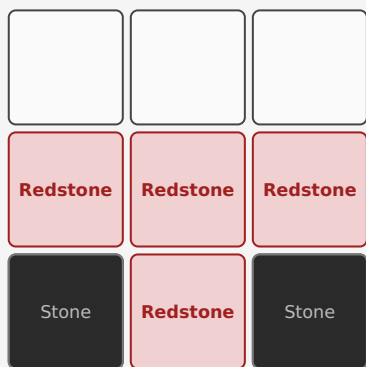
Yield: 1 dropper

Cobble	Cobble	Cobble
Cobble	Redstone	Cobble
Cobble	Redstone	Cobble

Tip: Same shape as dispenser but drops items instead of activating them. Useful for moving items between containers, dropping items into water streams, or ejecting items.

Redstone Repeater

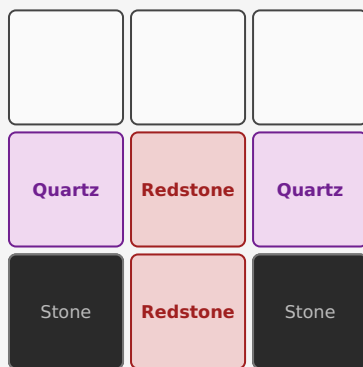
Yield: 1 repeater



Tip: Extends redstone signal up to 15 blocks, delays signal timing (right-click to cycle 1-4 ticks), and only passes signals from back to front (acts as a diode).

Redstone Comparator

Yield: 1 comparator



Tip: Has three modes: compare (outputs signal only if input is above reference), subtract (outputs difference), and measure container fullness (when facing a chest/hopper/furnace).

Redstone Block

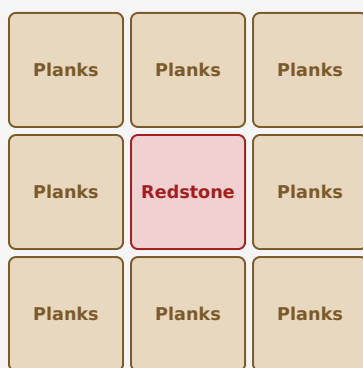
Yield: 1 redstone block



Tip: A permanent redstone power source (9 redstone dust compacted). Can be pushed by pistons. Also used to power a beacon.

Note Block

Yield: 1 note block



Tip: Plays a note when powered or clicked. Right-click to tune it. Different blocks underneath change the instrument (wood=bass, stone=snare, glass=hats, etc.).

6. Transportation

Minecart

Yield: 1 minecart

Iron		Iron
Iron	Iron	Iron

Tip: Ride in minecarts on rails. Right-click with a chest, hopper, furnace, or TNT in hand to create specialized minecarts.

Rail

Yield: 16 rails

Iron		Iron
Iron	Stick	Iron

Tip: The basic rail. Minecarts move slowly on flat rails. Use Powered Rails for acceleration.

Powered Rail

Yield: 6 powered rails

Gold	Stick	Gold
Gold	Redstone	Gold
Gold	Stick	Gold

Tip: Accelerates minecarts. Must be powered by redstone (lever, redstone torch, detector rail). Place one every 32 blocks to maintain speed.

Detector Rail

Yield: 6 detector rails

Iron	Redstone	Iron
Iron	Stone	Iron

Tip: The center-bottom is a Stone Pressure Plate. Emits redstone power when a minecart passes over. Good for station stops, automatic sorting, and alarms.

Boat

Yield: 1 boat

Planks		Planks
Planks	Planks	Planks

Tip: Fast travel on water. Right-click to enter, use WASD to steer. Also works on land as a temporary “sled” downhill. Add a chest for a Boat with Chest (carries items).

7. Enchanting & Upgrades

Enchanting Table

Yield: 1 enchanting table

	Diamond	
Obsidian	Obsidian	Obsidian
Obsidian	Book	Obsidian

Book recipe: 3 Paper in a V-shape + 1 Leather anywhere in the 2x2 grid. **Tip:** Surround the table with bookshelves (15 max) for level 30 enchantments. Leave 1 block of air between shelf and table.

Anvil

Yield: 1 anvil

Iron Block	Iron Block	Iron Block
	Iron	
Iron	Iron	Iron

Important: The top row is 3 **Iron Blocks** (9 iron ingots each = 27 ingots total). The bottom row is 4 **Iron Ingots**. Total: 31 iron. Used to repair items, combine enchantments, and rename items.

Bookshelf

Yield: 1 bookshelf

Planks	Planks	Planks
Book	Book	Book
Planks	Planks	Planks

Tip: Place around an enchanting table (1 block away) to increase enchantment levels. You need 15 bookshelves for max level 30 enchants.

Grindstone

Yield: 1 grindstone

Stick	Stone	Stick
	Planks	
	Planks	

Tip: Repairs items and removes enchantments. Disenchanted items return some XP. Also functions as a “weapon smith” job block for villagers.

8. Food & Consumables

Golden Apple

Yield: 1 golden apple

Gold Block	Gold Block	Gold Block
Gold Block	Apple	Gold Block
Gold Block	Gold Block	Gold Block

Cost: 8 **Gold Blocks** (72 gold ingots!) + 1 Apple. **Note:** This is the base golden apple — grants Absorption II + Regen II. The “Enchanted Golden Apple” (Notch Apple, with 8 gold *blocks*) is no longer craftable in newer versions.

Golden Carrot

Yield: 1 golden carrot

	Gold Ingot	
	Carrot	

Tip: Just 8 gold nuggets around a carrot in newer versions. Best saturation food in the game. Also used to breed horses and make night vision potions.

Bread

Yield: 1 bread

Wheat	Wheat	Wheat

Tip: Reliable food source from wheat farming. Three wheat in a row makes one bread. Easy to mass-produce once you have a farm set up.

Cake

Yield: 1 cake

Milk	Milk	Milk
Sugar	Egg	Sugar
Wheat	Wheat	Wheat

Tip: Place on ground and right-click to eat a slice (7 slices total). Milk is obtained by right-clicking a cow with a bucket. Cake does not grant saturation — better for decoration than survival.

Cookies

Yield: 8 cookies

Wheat	Cocoa	Wheat

Tip: Cocoa beans are found in jungle biomes on cocoa pods (jungle tree logs). Can be farmed. Cookies restore very little hunger but stack to 64.

Pumpkin Pie

Yield: 1 pumpkin pie

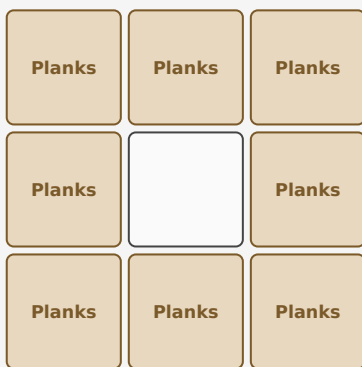
Pumpkin	Sugar	Egg

Tip: A top-tier food source. Easy to mass-produce if you have pumpkin and sugar cane farms plus a chicken for eggs.

9. Storage & Building Blocks

Chest

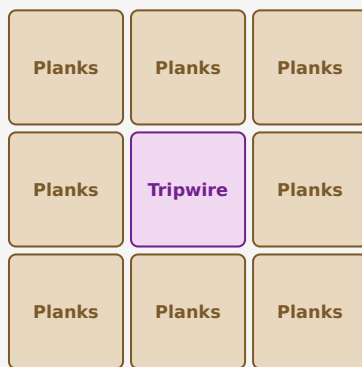
Yield: 1 chest



Tip: 27 inventory slots. Place two chests next to each other to make a **Large Chest** (54 slots). Cannot be placed directly next to another chest unless “locked” with a trapdoor in between.

Trapped Chest

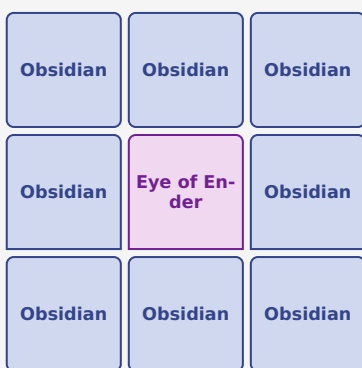
Yield: 1 trapped chest



Tip: Same storage as chest but emits a redstone signal when opened. Use for trap bases — hook it to a hidden dispenser or piston trap.

Ender Chest

Yield: 1 ender chest



Tip: Shared inventory across ALL ender chests you place. Perfect for moving items between bases. Only you can see your items. If broken, it drops as obsidian (lose the eye).

Furnace

Yield: 1 furnace



Tip: 8 cobblestone in a ring. Smelts ores, cooks food, makes glass and bricks. Use coal, charcoal, or wood planks as fuel. See the “Smelting” section below.

Crafting Table

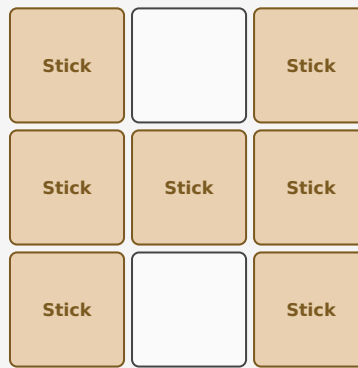
Yield: 1 crafting table



Tip: This is a 2x2 recipe in your personal inventory grid. Unlocks the full 3x3 grid. First thing to make in any survival world.

Ladder

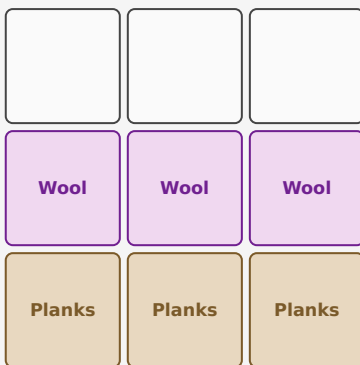
Yield: 3 ladders



Tip: Climb vertically. Place on the side of blocks. Easy to make and stackable — great for mineshafts and tall builds.

Bed

Yield: 1 bed



Tip: Right-click to sleep and set your spawn point (all players in a server must sleep). **WARNING:** Beds explode in the Nether and End — used as a weapon against the Ender Dragon!

Door (Wooden or Iron)

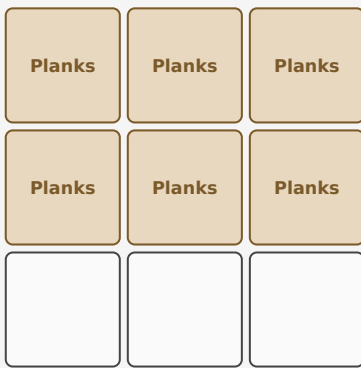
Yield: 3 doors



Tip: 6 planks = 3 wooden doors. Iron doors use iron ingots instead. Iron doors only open with redstone (button, lever, pressure plate). Mobs can open wooden doors but not iron ones.

Trapdoor

Yield: 2 trapdoors



Tip: Place in the top or bottom of a block space. Opens/closes with right-click or redstone. Good for hidden entrances, windows, and mob-proof fencing.

10. Smelting — Furnace Recipes

How the furnace works: Place the ingredient in the top slot, fuel (coal, charcoal, wood, lava bucket, etc.) in the bottom slot, and wait. Each operation takes about 10 seconds. A single piece of coal smelts 8 items. Multiple furnaces can run at once for faster processing.

Ore Smelting

Iron Ingot

Raw Iron (Ore)

↓ + Fuel ↓

Iron Ingot

Most important smelting. Needed for tools, armor, buckets, anvils, rails, and more. Mine iron ore with a stone pickaxe or better.

Gold Ingot

Raw Gold (Ore)

↓ + Fuel ↓

Gold Ingot

Used for golden apples, powered rails, clocks, and piglin bartering. Mine with an iron pickaxe or better.

Copper Ingot

Raw Copper (Ore)

↓ + Fuel ↓

Copper Ingot

Used for lightning rods, spyglasses, and building (copper oxidizes green over time). Mine with a stone pickaxe or better.

Material Smelting

Glass

Sand (or Red Sand)

↓ + Fuel ↓

Glass Block

Transparent building block.
Used for windows, greenhouses, beacons, and potion brewing (glass bottles).

Smooth Stone

Cobblestone

↓ + Fuel ↓

Stone

First smelt cobblestone into regular stone, then smelt stone into smooth stone. Used for blast furnaces and building. Run stone twice through the furnace for smooth stone.

Brick

Clay Ball

↓ + Fuel ↓

Brick (item)

Clay is found underwater (lakes, rivers, oceans). Break clay blocks for 4 clay balls each. Bricks are crafted 2×2 into Brick Blocks for building.

Nether Brick

Netherrack

↓ + Fuel ↓

Nether Brick (item, via crafting)

Actually, nether bricks are crafted from nether wart blocks found in fortresses, not smelted. Smelt netherrack for **Nether Quartz** or just mine it.

Terracotta

Clay Block

↓ + Fuel ↓

Terracotta

Durable decorative block. Comes in 16 colors. Dye terracotta to make glazed terracotta for unique patterns.

Charcoal

Wood Log

↓ + Fuel ↓

Charcoal

A renewable alternative to coal. Smelt logs using any fuel (including other charcoal). Each charcoal smelts 8 items — same as coal. Essential if you can't find coal.

Dried Kelp

Kelp

↓ + Fuel ↓

Dried Kelp

Food that can be eaten or crafted into Dried Kelp Blocks (compact fuel). Each block smelts 20 items — better than coal! Farm kelp underwater for infinite fuel.

Food Cooking

Steak

Raw Beef

↓ + Fuel ↓

Cooked

One of the best food sources.
Farm cows for reliable food.

Cooked Porkchop

Raw Porkchop

↓ + Fuel ↓

Cooked Porkchop

Equal to steak in hunger
restored. Pigs drop 1-3 raw
porkchops.

Cooked Chicken

Raw Chicken

↓ + Fuel ↓

Cooked Chicken

Eating raw chicken has a
30% chance of food
poisoning (hunger effect).
Always cook your chicken.

Cooked Cod/Salmon

Raw Fish

↓ + Fuel ↓

Cooked Fish

Easy early food from fishing
or killing fish. Also feeds cats
and tames ocelots.

Baked Potato

Potato

↓ + Fuel ↓

Baked Potato

Excellent food source.
Potatoes are easy to farm.
Eating baked potatoes
restores 5 hunger points —
very efficient.

Cooked Mutton

Raw Mutton

↓ + Fuel ↓

Cooked Mutton

Sheep drop mutton. Good
food if you have a sheep
farm (also get wool).

Cooked Rabbit

Raw Rabbit

↓ + Fuel ↓

Cooked Rabbit

Rabbits are found in deserts,
flower forests, and taigas.
Also used in Rabbit Stew
(best stew in the game).

11. Advanced Machines

Blast Furnace

Yield: 1 blast furnace

Iron	Iron	Iron
Iron	Furnace	Iron
Smooth Stone	Smooth Stone	Smooth Stone

Tip: Smelts ore items (iron, gold, etc.) at **double speed** compared to a regular furnace. Uses the same fuel. Also serves as an armorer villager job block.

Smoker

Yield: 1 smoker

Log	Log	Log
	Furnace	

Tip: Cooks food at **double speed** compared to a regular furnace. Same fuel efficiency but half the time. Essential kitchen block.

Stonecutter

Yield: 1 stonecutter

Iron	Stone	

Tip: Craft stone blocks (stairs, slabs, walls) with 1:1 ratio instead of losing materials in the crafting table. Also a stone mason villager job block.

Brewing Stand

Yield: 1 brewing stand

	Blaze Rod	
Cobble	Cobble	Cobble

Tip: Used to brew potions. Blaze Rods are obtained by killing Blazes in the Nether. Three cobblestone on the bottom, one blaze rod on top.

12. Nether Portal

Building a Nether Portal

Requires: 10 to 14 Obsidian + Flint and Steel



How to get obsidian: Pour a water bucket onto lava source blocks (not flowing lava). Mine with a **diamond pickaxe**. You need at least 10 obsidian for a minimum portal.

Shape: Minimum 4 blocks wide × 5 blocks tall (inside dimensions). Max 23×23. Light the inside bottom block with Flint and Steel.

Tip: You can save obsidian by leaving the corners out (needs only 10 instead of 14). The corners of the frame don't need to be obsidian.

Nether Portal - Quick Facts:

- Must be a vertical rectangle, at least 4×5 (inside)
- Corners are optional — save 4 obsidian
- Light with Flint and Steel on the bottom block inside
- You can place water inside to stop pigmen from coming through (Java edition)
- Coordinates in Nether correspond ~1:8 to Overworld
- Destroy the portal by breaking any obsidian frame block or placing water on the activated portal

13. Block Compression (Storage)

Diamond Block

Yield: 1 diamond block (9 diamonds)

Diamond	Diamond	Diamond
Diamond	Diamond	Diamond
Diamond	Diamond	Diamond

Tip: Compact storage — 9 diamonds = 1 block. Use as beacon base, decoration, or flex. Can be uncrafted back into 9 diamonds.

Iron Block

Yield: 1 iron block (9 iron ingots)

Iron	Iron	Iron
Iron	Iron	Iron
Iron	Iron	Iron

Tip: Used for Anvil top row (3 blocks = 27 ingots), Iron Golem builds, and beacon pyramids. Beacons need up to 164 blocks total for max level.

Gold Block

Yield: 1 gold block (9 gold ingots)

Gold	Gold	Gold
Gold	Gold	Gold
Gold	Gold	Gold

Tip: Used for Golden Apple (8 blocks = 72 ingots + apple), beacon bases, and piglin bartering (piglins are attracted to gold blocks).

Copper Block

Yield: 1 copper block (9 copper ingots)

Copper	Copper	Copper
Copper	Copper	Copper
Copper	Copper	Copper

Tip: Copper oxidizes over time (goes from orange → brown → green). Can be waxed to preserve a stage. Used in lightning rods and spyglasses.

Dried Kelp Block

Yield: 1 dried kelp block

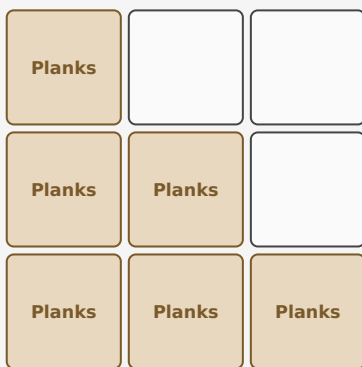
Dried Kelp	Dried Kelp	Dried Kelp
Dried Kelp	Dried Kelp	Dried Kelp
Dried Kelp	Dried Kelp	Dried Kelp

Best fuel: Smelts 20 items per block — beats coal (8 items). Create an underwater kelp farm for infinite green fuel.

14. Decorative and Essential Blocks

Stairs

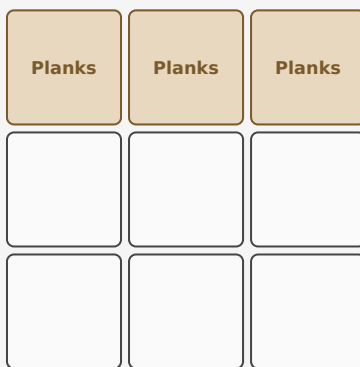
Yield: 4 stairs



Tip: Works with any full block material (stone, wood, etc.). Jumping up stairs uses less hunger than jumping up full blocks.

Slab (x6)

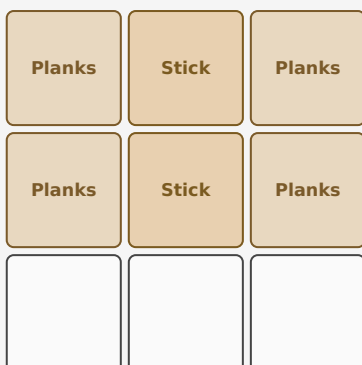
Yield: 6 slabs



Tip: Place in the top/bottom half of a block. Mobs cannot spawn on bottom-half slabs. Useful for mob-proof flooring.

Fence

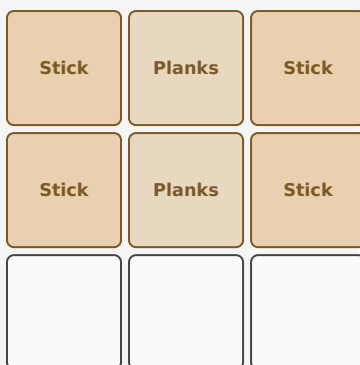
Yield: 3 fences



Tip: Mobs cannot jump over fences. 1.5 blocks tall. Connect to other fences automatically. Combine with fence gates for entry.

Fence Gate

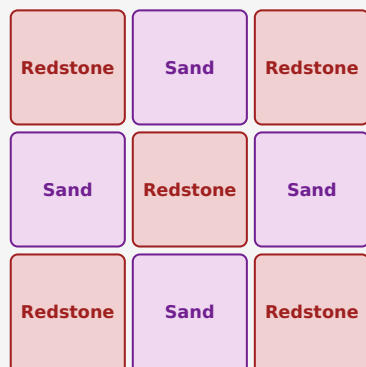
Yield: 1 fence gate



Tip: Opens and closes like a door but matches fence height. Right-click to open/close. Also works with redstone.

TNT

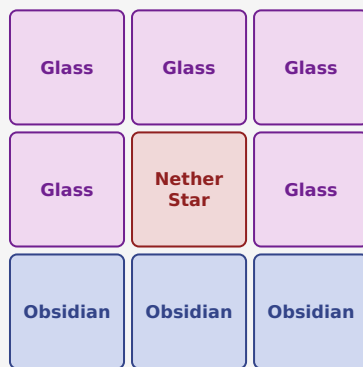
Yield: 1 TNT



Tip: Explodes when powered by redstone, shot with a flaming arrow, or hit by any explosion. Checkerboard pattern: 4 sand + 5 redstone dust. Use for mining, traps, or fun.

Beacon

Yield: 1 beacon



Tip: Place on top of a pyramid of iron/gold/diamond/emerald/copper/netherite blocks. Grants status effects (haste, speed, jump boost, resistance, strength, regeneration) to nearby players.

15. Quick Reference: Crafting Shape Patterns

Memorize these patterns — they stay the same across all material tiers. Just swap the material (Wood, Stone, Iron, Gold, Diamond, Netherite).

Item	Pattern (top row → middle → bottom)	Materials
Pickaxe	MMM / _S_ / _S_	3 material + 2 sticks
Axe	MM_ / MS_ / _S_	3 material + 2 sticks
Sword	_M_ / _M_ / _S_	2 material + 1 stick
Shovel	_M_ / _S_ / _S_	1 material + 2 sticks
Hoe	MM_ / _S_ / _S_	2 material + 2 sticks
Helmet	MMM / M_M	5 material
Chestplate	M_M / MMM / MMM	8 material
Leggings	MMM / _M_ / M_M	7 material
Boots	_ _ / M_M / M_M	4 material
Block (compressed)	MMM / MMM / MMM	9 ingots/gems
Shield	PIP / PPP	6 planks + 1 iron ingot
Bucket	_ _ / I_I / _I_	3 iron ingots
Compass	_I_ / IRI / _I_	4 iron + 1 redstone dust
Piston	PPP / CIC / CRC	3 planks + 4 cobble + 1 iron + 1 redstone
Door (3)	MMM / MMM / MMM	6 planks or 6 iron ingots
Trapdoor (2)	MMM / MMM	6 planks
Fence (3)	MSM / MSM	4 planks + 2 sticks
Stairs (4)	M__ / MM_ / MMM	6 material blocks
Slab (6)	MMM	3 material blocks

Legend: M = Material (Wood/Stone/Iron/Gold/Diamond/Netherite/Planks/Stone types) • S = Stick • P = Planks • I = Iron • R = Redstone • C = Cobblestone • _ = Empty. Patterns are shown from left to right, top to bottom.

About this guide: Version 1.1 — Updated with full item names, smelting recipes, expanded building blocks, food items, and clearer notation. All recipes are for Minecraft Java/Bedrock Edition. No chemical symbols used — everything labeled with its in-game name. Free & open source.